Aeden Jameson
Marcus Deichman

MiniProject 3 Kaggle Report

Our best model evaluated the Kaggle test dataset with an accuracy of 94.8% using a zero-shot prompt. On the internal test dataset accuracy was 99%. Below are the experimentation details:

The Model: Llama 3.2 1B

- The following Lora config was constant through out:
    - task_type="CAUSAL_LM",bias="none",
- The following training config was constant through out:
    - batch_size=4, gradient_accumulation=8

	Lora & Training Parameters	**Acc**
Exp 1	r=8, lora_alpha=16, target_modules=['q_proj', 'v_proj'],lora_dropout=0.01, epochs=3, lr=2e-5, decay=0.01	*18%*
Exp 2	Same as Exp 1, epochs=10	*84%*
Exp 3	Same as Exp 2, target_modules=['q_proj', 'v_proj', 'k_proj'],	*89%*
Exp 4	Same as Exp 1, target_modules='all-linear'	*81%*
Exp 5	Same as Exp 4, epochs=10	*98%*
Exp 6	Same as Exp 5, r=16, lora_alpha=32	*99%*
Exp 7 **Best**	Same as Exp 6, LoRa Dropout=0.10, epochs=15 Kaggle Accuracy: **94.8%**	*99%*

Observations

- Our best configurations converged at around 10-12 epochs. When going to 20 epochs, the accuracy improved on the local test dataset but performed around 0.1% worse on the Kaggle dataset.
- The number of epochs and trainable parameters had the biggest impact on accuracy and F1-scores.

- Second, rank had the next biggest impact on classification
- Upping the dropout rate provided slightly improved results when training at larger epochs since it reduced the risk of overfitting. Further experiments are needed if a higher dropout rate with larger epochs can further improve results.